INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Digital Realty CH1, IL -- station KORD, America/Chicago
Committed pair	OSM 377032061 -> 377032058 Equinix Chicago CH3 / EdgeConneX CHI02
Facade gap	136.2 m between the two halls
Weather record	43,775 real hourly records from KORD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3191 C at 265 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2025-08-30 (recirc_edge)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 2 mode changes. That is 1 more than the reactive incumbent, at 0 unsafe hours against its 0. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	8 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	18.213	21.0	17.780	1.069
01	FREE	yes	-	18.196	21.0	17.780	1.004
02	FREE	yes	-	18.205	21.0	17.780	0.976
03	FREE	yes	-	18.139	21.0	17.780	0.846
04	FREE	yes	-	18.248	21.0	17.780	0.896
05	FREE	yes	-	18.282	21.0	17.780	0.812
06	FREE	yes	-	18.719	21.0	18.330	0.765
07	FREE	yes	-	19.899	21.0	18.330	1.401
08	mechanical	no	dry-bulb	21.198	21.0	18.890	1.752
09	mechanical	no	dry-bulb	22.363	21.0	19.440	1.853
10	mechanical	no	dry-bulb	22.907	21.0	20.560	1.721
11	mechanical	no	dry-bulb	22.856	21.0	21.110	1.306

12	mechanical	no	dry-bulb	23.133	21.0	21.670	1.341
13	mechanical	no	dry-bulb	23.863	21.0	22.529	1.274
14	mechanical	no	dry-bulb	23.221	21.0	22.220	1.012
15	mechanical	no	dry-bulb	22.955	21.0	21.670	1.095
16	mechanical	no	dry-bulb	22.597	21.0	21.110	1.109
17	mechanical	no	dry-bulb	21.765	21.0	20.560	0.925
18	mechanical	yes	switch budget	20.914	21.0	20.000	0.944
19	mechanical	yes	switch budget	20.104	21.0	18.890	1.180
20	mechanical	yes	switch budget	19.561	21.0	18.330	1.212
21	mechanical	yes	switch budget	18.977	21.0	17.780	1.051
22	mechanical	yes	switch budget	18.439	21.0	17.780	0.904
23	mechanical	yes	switch budget	18.201	21.0	17.220	1.103

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.213 C, which is 2.787 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.069 C, and the margin is measured, not chosen: 0.912 C of group-conditional forecast error for this hour of day, plus 0.1578 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.196 C, which is 2.804 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.004 C, and the margin is measured, not chosen: 0.863 C of group-conditional forecast error for this hour of day, plus 0.1412 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.205 C, which is 2.795 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.976 C, and the margin is measured, not chosen: 0.825 C of group-conditional forecast error for this hour of day, plus 0.1502 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.139 C, which is 2.861 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.846 C, and the margin is measured, not chosen: 0.762 C of group-conditional forecast error for this hour of day, plus 0.0838 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.248 C, which is 2.752 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.896 C, and the margin is measured, not chosen: 0.710 C of group-conditional forecast error for this hour of day, plus 0.1861 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.282 C, which is 2.718 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.812 C, and the margin is measured, not chosen: 0.615 C of group-conditional forecast error for this hour of day, plus 0.1973 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.719 C, which is 2.281 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.765 C, and the margin is measured, not chosen: 0.661 C of group-conditional forecast error for this hour of day, plus 0.1043 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.899 C, which is 1.101 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.401 C, and the margin is measured, not chosen: 1.225 C of group-conditional forecast error for this hour of day, plus 0.1755 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.198 C against a 21.0 C limit, so it fails by 0.198 C. It would take a limit 0.198 C higher, or a bound 0.198 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.363 C against a 21.0 C limit, so it fails by 1.363 C. It would take a limit 1.363 C higher, or a bound 1.363 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.907 C against a 21.0 C limit, so it fails by 1.907 C. It would take a limit 1.907 C higher, or a bound 1.907 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.856 C against a 21.0 C limit, so it fails by 1.856 C. It would take a limit 1.856 C higher, or a bound 1.856 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.133 C against a 21.0 C limit, so it fails by 2.133 C. It would take a limit 2.133 C higher, or a bound 2.133 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.863 C against a 21.0 C limit, so it fails by 2.863 C. It would take a limit 2.863 C higher, or a bound 2.863 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.221 C against a 21.0 C limit, so it fails by 2.221 C. It would take a limit 2.221 C higher, or a bound 2.221 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.955 C against a 21.0 C limit, so it fails by 1.955 C. It would take a limit 1.955 C higher, or a bound 1.955 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.597 C against a 21.0 C limit, so it fails by 1.597 C. It would take a limit 1.597 C higher, or a bound 1.597 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.765 C against a 21.0 C limit, so it fails by 0.765 C. It would take a limit 0.765 C higher, or a bound 0.765 C tighter, to change this hour.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.914 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.104 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.561 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.977 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.439 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.200 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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